MIL-S-61B
21 OCTOBER 1959
SUPERSEDING
MIL-S-61A
15 SEPTEMBER 1952

MILITARY SPECIFICATION

SHUNTS, INSTRUMENT, EXTERNAL, 50 MILLIVOLT (LIGHTWEIGHT TYPE)

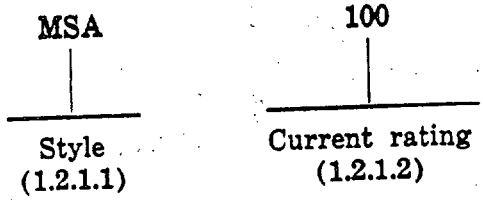
This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers a lightweight type of external instrument shunt rated at 50 millivolts (mv) in current ratings from 10 to 1,200 amperes, inclusive (see 6.1).

1.2 Classification.

1.2.1 *Type designation.* The type designation shall be in the following form, and as specified (see 3.1 and 6.2):



1.2.1.1 *Style.* The style is identified by the three-letter symbol. The letters "MS" identify 50 mv, external instrument shunts, and the letter "A" identifies the shape and dimensions of the shunt.

1.2.1.2 *Current rating.* The current rating expressed in amperes is identified by a three-digit number; the first two digits represent significant figures and the last digit specifies the number of zeros to follow.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

FF-W-92	— Washers, Metal, Flat Plain.
QQ-B-613	— Brass, Leaded and Nonleaded; Plate, Rolled-Bar, Sheet and Strip.
QQ-B-626	— Brass, Leaded and Nonleaded; Rods, Shapes, Forgings and Flat Products with Finished Edges (Bars, Flat Wire, and Strips).
QQ-S-561	— Solder; Silver.

FSC 6625

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QQ-S-571 — Solder: Lead Alloy, Tin Lead Alloy, and Tin Alloy; Flux Cored Ribbon and Wire and Solid Form.

PPP-B-566 — Boxes, Folding, Paperboard.

PPP-B-585 — Boxes; Wood, Wirebound.

PPP-B-591 — Boxes, Fiberboard, Wood-Cleated.

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Boxes Fiber.

PPP-B-665 — Boxes, Paperboard, Metal Stayed (Including Stay Material).

PPP-B-676 — Boxes, Setup, Paperboard.

PPP-T-76 — Tape; Pressure-Sensitive Adhesive, Paper, Water-Resistant.

PPP-T-97 — Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-M-14 — Molding Plastics and Molded Plastic Parts, Thermosetting.

MIL-P-116 — Preservation, Methods of.

MIL-B-857 — Bolts, Nuts, and Studs.

MIL-S-933 — Screws, Machine, Cap and Set; and Nuts.

MIL-S-6872 — Soldering Process General Specification for.

MIL-B-10377 — Box, Wood, Cleated, Veneer, Paper-Overlaid.

MIL-L-10547 — Liners, Case, Waterproof.

MIL-P-15035 — Plastic-Material, Laminated, Thermosetting: Sheets, Cotton--Fabric-Base, Phenolic-Resin.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storag

MIL-STD-130 — Identification Marking of U. S. Military Property.

MIL-STD-202 — Tests Methods for Electronic and Electrical Component Parts.

MS15795 — Washers, Flat, Metal, Round, General Purpose.

MS35229 — Screw, Machine, Pan Head, Slotted, Brass, Plain Finish, NC-2A and UNC-2A.

MS35338 — Washer, Lock, Split, Helical, Medium Series.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting office

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2.2 **Other publications.** The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 — Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, One Park Avenue, at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Detail requirements for individual shunt styles. Detail requirements or exceptions applicable to individual styles of shunts shall be as specified on figures 1 to 3, inclusive. In the event of any conflict between requirements of this specification and the detail requirements, the latter shall govern (see 6.2).

3.2 Material. The material for each part shall be as specified herein. However, when a definite material is not specified, a material shall be used which enable the shunts to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.2.1 *Plastic (thermosetting).*

3.2.1.1 *Molding.* Molding plastic shall conform to type CFI–10, or higher impact-strength type, of Specification MIL–M–14.

3.2.1.2 *Laminated.* Laminated plastic shall conform to Specification MIL–P–15035.

3.2.2 *Metals.* The use of ferrous metals is not permitted.

3.2.2.1 *Solder.* Solder used for electrical connections shall conform to Specification QQ–S–561 or QQ–S–571, and shall be of a composition which will not flow under the conditions specified herein.

3.2.2.2 *Threaded parts and washers.* Unless otherwise specified (see 3.1), machine screws, capscrews, nuts, studs, and plain washers shall be made of brass conforming to Specifications QQ–B–613 and QQ–B–626.

3.2.2.3 *Shunt leaf.* Material for the shunt leaf shall be shunt manganin or an alloy having an equivalent temperature coefficient of resistance at any temperature within the working range of the shunt.

3.3 Design and construction. Shunts shall be of the design, construction, and physical dimensions specified (see 3.1).

3.3.1 *Threaded parts.* Unless otherwise specified (see 3.1), all threaded parts shall be in accordance with Handbook H28. Where practical, all threads shall be in conformity with the coarse-thread series. The fine-thread series shall be used only for applications that might show a definite advantage through their use. Where a special diameter-pitch combination is required, the thread shall be of American National Form and of any pitch between 16 and 36 which is used in the fine-thread series.

3.3.1.1 *Engagement of threaded parts.* All threaded parts shall engage by at least four full threads.

3.3.1.2 *Machine screws capscrews, nuts, and studs.* Unless otherwise specified (see 3.1), all machine screws, capscrews, nuts, and studs shall conform to Specification MIL–B–857 or MIL–S–933, as applicable.

3.3.2 *Plain washers.* Unless otherwise specified (see 3.1), plan washers shall conform to Specification FF–W–92.

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3.3.3 *Shunt base.* The shunt shall be mounted on a molded- or laminated-plastic base of the dimensions specified (see 3.1). When a molded-plastic base is used, either a channel or two U-shaped cuts in the base shall be permitted. When a laminated-plastic base is used, either the U-shaped cuts or a counterbore of the dimensions specified (see 3.1) shall be permitted. The base shall be securely attached to the shunt-terminal blocks with screws of adequate size, and shall be made resistant to loosening under the conditions specified herein. The heads of the screws shall be recessed in the base and properly protected with a good grade of insulating compound which shall not flow, crack, chip, pulverize, or otherwise deteriorate substantially under the conditions specified herein.

3.4 Dielectric withstanding voltage. When shunts are tested as specified in 4.6.2, there shall be no evidence of visible damage, arcing, or breakdown.

3.5 Voltage drop. The voltage drop under rated current (see 3.1) shall be 50 ± 0.3 mv (see 4.6.3).

3.6 Sustained load. When shunts are tested as specified in 4.6.4, the permanent change in voltage drop shall not exceed 0.5 percent of the initial value.

3.7 Overload cycling. When shunts are tested as specified in 4.6.5, the permanent change in voltage drop shall not exceed 0.5 percent of the initial value. There shall be no evidence of deformation loosening of the base, or other visible damage to the shunt.

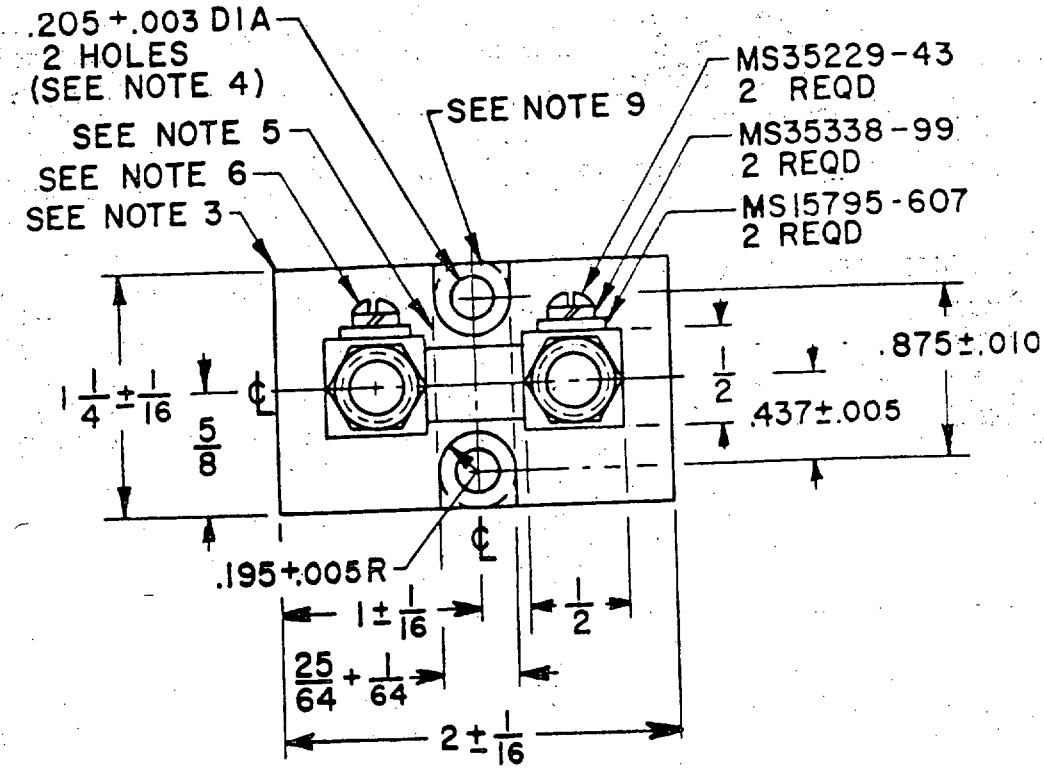
3.8 Shock. When shunts are tested as specified in 4.6.6, the permanent change in voltage drop shall not exced 0.5 percent of the initial value. There shall be no evidence of deformation, loosening of the base, or other visible damage to the shunt.

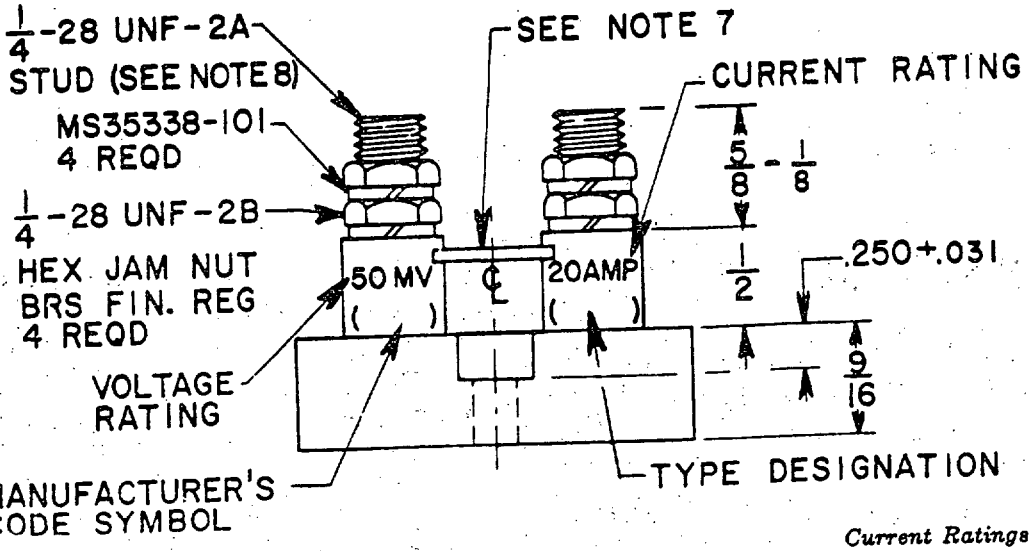
3.9 Marking. Shunts shall be marked in accordance with Standard MIL-STD-130, with the type designation, the manufacturer's code symbol, the nominal voltage rating, the current rating, and other identification marking as specified (see 6.2). There shall be no space between the symbols which comprise the type designation. The marking shall appear in the designated spaces and in the form specified (see 3.1).

3.10 Workmanship. Shunts shall be manufactured and processed in a careful and workmanlike manner, in accordance with good design and sound practice. Shunts shall be uniform, clean, and free from burrs, cracks, surface roughness, and any other evidence of poor quality and workmanship which is such that the shunts are not suitable for the purpose intended.

3.10.1 *Soldering.* When corrosive fluxes are employed in soldering, the corrosive elements, excess flux, and solder shall be removed after soldering. Where applicable, soldering shall be in accordance with Specification MIL-S-6872.

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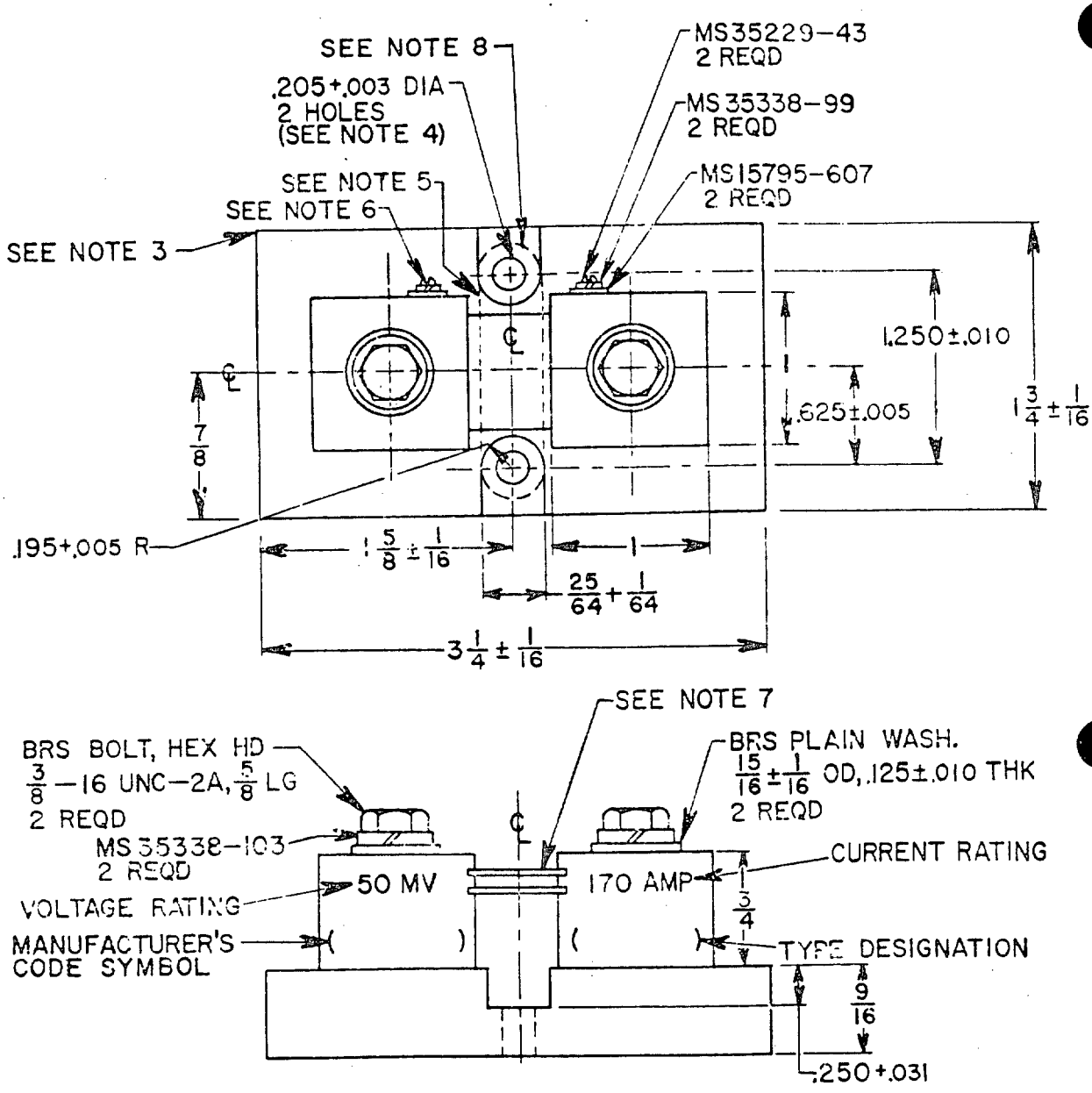
1. All dimensions in inches.
2. Unles otherwise specified, tolerances are ± 1/32 on fractions.
3. Corners may be square or have radius of 0.125 maximum.
4. Holes shall be centered in channel, U-cut, or counterbore.
5. For molded-plastic base, channel optional.
6. Terminal position optional.
7. Number of leaves optional.
8. Metal for the studs shall be either brass or copper.
9. A counterbore of 0.390 + 0.010 diameter shall be permitted, as an alternate construction for laminated-plastic bases.

FIGURE 1. *Style MSA shunt.*

Current Ratings

Type designation	Current rating (amperes)
MSA100	10
MSA150	15
MSA200	20
MSA300	30
MSA500	50
MSA750	75
MSA800	80
MSA850	85
MSA101	100
MSA151	150

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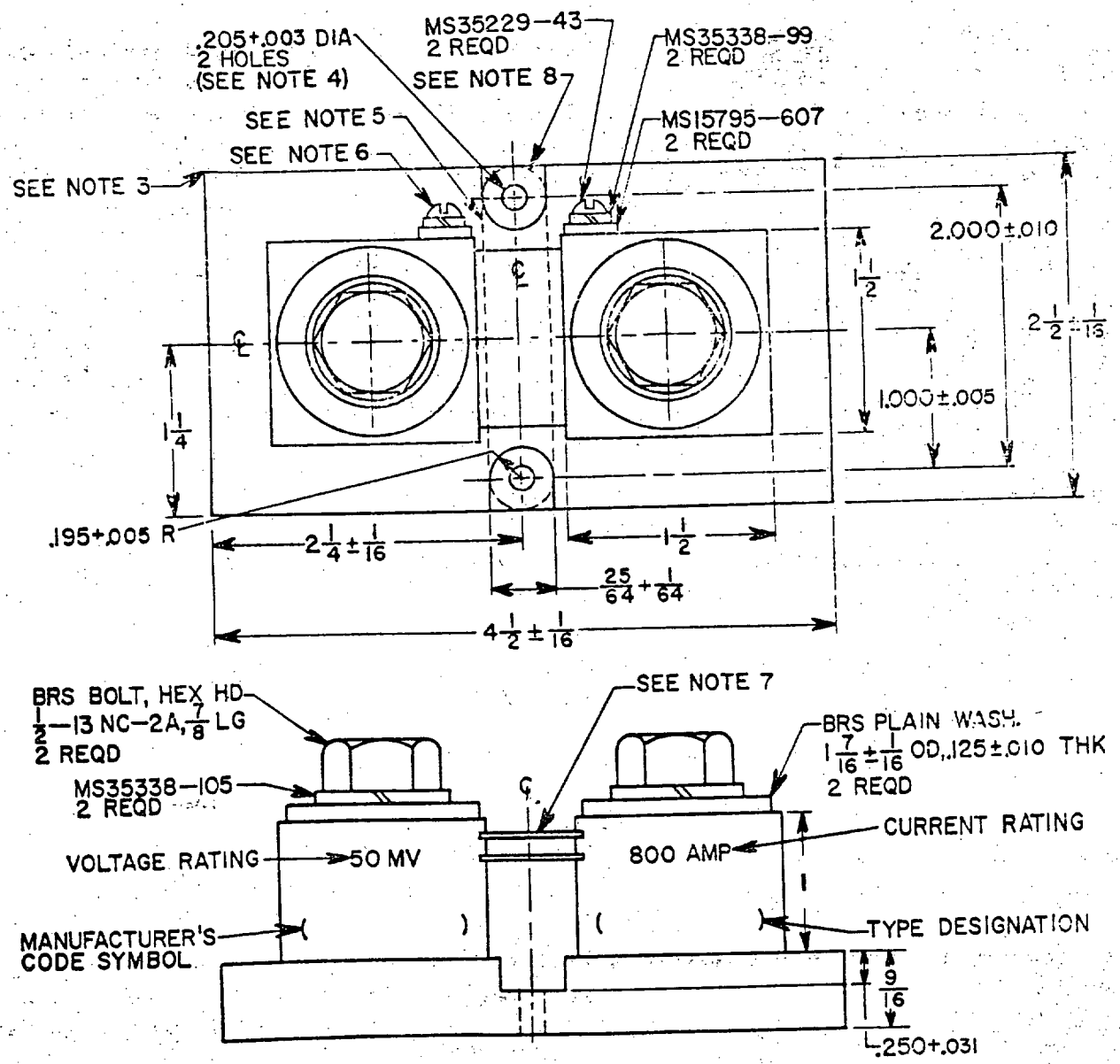
1. All dimensions in inches.
2. Unless otherwise specified, tolerances are ± 1/32 on fractions.
3. Corners may be square or have radius of 0.125 maximum.
4. Holes shall be centered in channel, U-cut, or counterbore.
5. For molded-plastic base, channel optional.
6. Terminal position optional.
7. Number of leaves optional.
8. A counterbore of 0.390 + 0.010 diameter shall be permitted, as an alternate construction for laminated-plastic bases.

Current Ratings

Type designation	Current rating (amperes)
MSB171	170
MSB201	200
MSB251	250
MSB301	300
MSB401	400
MSB451	450
MSB501	500
MSB601	600

FIGURE 2. *Style MSB shunt.*

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1. All dimensions in inches.
2. Unless otherwise specified, tolerances are ± 1/32 on fractions.
3. Corners may be square or have radius of 0.125 maximum.
4. Holes shall be centered in channel, U-cut, or counterbore.
5. For molded-plastic base, channel optional.
6. Terminal position optional.
7. Number of leaves optional.
8. A counterbore of 0.390 + 0.010 diameter shall be permitted, as an alternate construction for laminated-plastic bases.

Current Ratings

Type designation	Current rating (amperes)
MSC801	800
MSC102	1,000
MSC122	1,200

FIGURE 3. *Style MSC shunt.*

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 *Supplier.* Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Unless otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government, as specified in the contract or order.

4.1.1.1 *Test equipment and inspection facilities.* Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government.

4.1.2 *Government.* Acceptance of the shunts shall be based upon verification by the Government of the supplier's compliance with the requirements of this specification. The Government may, at its option, repeat any or all of the inspections specified herein.

4.2 Classification of inspection. The examination and testing of shunts shall be classified as follows:

4.2.1 *Acceptance inspection.*

4.2.1.1 *Inspection of product for delivery* (see 4.4).

4.2.1.2 *Inspection of preparation for delivery* (see 4.5).

4.3 Inspection conditions and methods. Unless otherwise specified herein all inspection shall be made at room ambient temperature, pressure, and humidity.

4.3.1 *Reference test temperature.* The reference test temperature of the shunts shall be 110° ± 5°C. as measured at the approximate center of the outer-shunt-blade surface. Shunt temperature shall be determined by the thermocouple method.

4.3.2 *Connections.* The current-carrying leads used with the shunt shall be of adequate size (see 6.3), and shall be provided with terminals of the straight, square-tongue type, with minimum tongue dimensions equal to those of the current-terminal face of the shunt block. All lead terminals shall extend away from the shunt-terminal block in a direction parallel to the length of the base.

4.4 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B.

4.4.1 *Inspection lot.* An inspection lot shall be as specified in Standard MIL–STD–105, and shall consist of all shunts of the same style and current rating.

4.4.2 *Resubmitted lots.* If an inspection lot is rejected, the manufacturer may rework the lot or screen out defectives and resubmit it for acceptance inspection. Resubmitted lots shall be kept separate from new lots. The resubmitted lot shall be inspected, using tightened inspection.

4.4.3 *Group A inspection.* Group A inspection shall consist of the examinations and tests specified in table I, in the order shown. Statistical sampling and inspection shall be in accordance with Standard MIL–STD–105. The acceptable quality levels (AQL) shall be as specified in table I. Major and minor defects shall be as defined in Standard MIL–STD–105.

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TABLE I. *Group A inspection*

Examination or test	Requirement paragraph	Method paragraph	AQL (percent defective)	
			Major	Minor
Visual and mechanical examination:	...	4.6.1	1.0	4.0
Material	3.1 to 3.2.2.3, incl.	...		
Threaded parts	3.3.1 to 3.3.2, incl.	...		
Mounting dimensions	3.3.3	...		
Other dimensions	3.3	...		
Shunt base	3.3.3	...		
Marking	3.9	...		
Workmanship	3.10 and 3.10.1	...		
Dielectric withstanding voltage	3.4	4.6.2	1.0	...
Voltage drop	3.5	4.6.3		

4.4.4 *Group B inspection.* Group B inspection shall consist of the tests specified in table I, in the order shown.

TABLE II. *Group B inspection*

Test	Requirement paragraph	Method paragraph
Sustained load	3.6	4.6.4
Overload cycling	3.7	4.6.5
Shock	3.8	4.6.6

4.4.4.1 *Sampling procedure.* The sampling procedure shall be in accordance with the appendix to Standard MIL–STD–105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R–1. The AQL shall be 4.0 (percent defective), and the inspection level shall be L6 for normal and tightened inspection and L4 for reduced inspection.

4.4.4.2 *Disposition of sample units.* Sample units which have been subjected to group B inspection shall not be delivered on the contract or order.

4.5 Inspection of preparation for delivery. Preparation for delivery inspection shall be in accordance with Specification MIL–P–116.

4.6 Method of examination and test.

4.6.1 *Visual and mechanical examination.* Shunts shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1 to 3.3.3, incl., and 3.9 to 3.10.1, incl.).

4.6.2 *Dielectric withstanding voltage (see 3.4).* Shunts shall be tested in accordance with method 301 of Standard MIL–STD–202. The following details and exceptions shall apply:

(a) Special preparation or conditions — Shunts shall be mounted on a flat metal plate, using a jig to simulate normal mounting conditions.

(b) Magnitude of test voltage — 1,500 volts root mean square of a commercial-line frequency.

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(c) Nature of potential — Alternating current.

(d) Points of application of test voltage — Between shunt-terminal block and the metal plate.

(e) Rate of application — The test voltage shall be applied gradually, increasing the voltage at an approximately uniform rate so that the specified voltage is reached in not less than 5 seconds nor more than 10 seconds.

(f) Examinations after dielectric-withstanding-voltage test — Shunts shall be visually examined for any evidence of damage, arcing or breakdown.

4.6.3 *Voltage drop.* The voltage drop in millivolts as measured between the voltage terminals of the shunt shall be ascertained under rated current (see 3.1) at the reference test temperature specified in 4.3.1. If the shunts do not reach the reference test temperature, any means may be used to attain this temperature. Measurement shall be made by the potentiometer method, or its equivalent (see 3.5).

4.6.4 *Sustained load.* Following a measurement of voltage drop as specified in 4.6.3, rated current (see 3.1) shall be passed through the shunt continuously for 2 hours. The temperature of the shunt-terminal blocks shall be maintained at approximately 145°C. during this period, using external means if necessary. Following a return to room ambient temperature, the voltage drop shall again be measured as specified in 4.6.3 (see 3.6).

4.6.5 *Overload cycling.* Following a measurement of voltage drop as specified in 4.6.3, sufficient current shall be passed through the shunt to cause the shunt-terminal-block temperature to reach 160°C. but not great enough to cause any part of the shunt blades to exceed 220°C. The current shall then be disconnected and the shunt cooled for a minimum of 1 hour at room ambient temperature. This cycle shall be repeated 10 times. At the end of the tenth cycle, the shunt shall be permitted to return to room ambient temperature and the voltage drop shall again be measured as specified in 4.6.3. The shunts shall then be examined for evidence of deformation, loosening of the base, or other visible damage (see 3.7).

4.6.6 *Shock (see 3.8).* Shunts shall be tested in accordance with method 202 of Standard MIL-STD-202. The following details shall apply:

(a) Mounting method — The shunt shall be securely mounted by its normal mounting means on the carriage of the shock-testing apparatus.

(b) Mounting accessories — None.

(c) Acceleration requirements — Fifty gravity units.

(d) Number of blows — Ten blows each in a direction parallel to each of the three principal axes (total of 30 blows).

(e) Measurements after shock — After the completion of the 30 blows, the voltage drop shall again be measured as specified in 4.6.3.

(f) Examinations after shock — The shunt shall be examined for evidence of deformation, loosening of the base, or other visible damage.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (see 6.2).

5.1.1 *Level A.* Unless otherwise specified (see 6.2), shunts shall be individually protected and unit-packaged in accordance with method IA of Specification MIL-P-116, without the use of contact preservatives. When method IA-8 is used, each packaged shunt shall be placed in an individual unit

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container conforming to Specification PPP–B–566, PPP–B–665, or PPP–B–676.

5.1.2 *Level C.* Shunts shall be afforded preservation and packaging in accordance with the manufacturer's normal commercial practice.

5.2 Packing (see 6.2).

5.2.1 *Level A.* Shunts packaged as specified (see 6.2) shall be packed in overseas-type wirebound wood, wood-cleated fiberboard, wood-cleated plywood, nailed wood, fiber (class 2 or 3, as specified (see 6.2)), or wood-cleated paper-overlaid boxes conforming to Specifications PPP–B–585, PPP–B–591, PPP–B–601, PPP–B–621, PPP–B–636, and MIL–B–10377, respectively, at the option of the manufacturer. Shipping containers shall have case liners conforming to Specification MIL–L–10547; the case liners shall be closed and sealed in accordance with the appendix thereto. Case liners for boxes conforming to Specification PPP–B–636 may be omitted provided the center and edge seams and manufacturer's joints are sealed with tape, at least 1½ inches wide, conforming to Specification PPP–T–76. Box closures shall be as specified in the applicable box specification or appendix thereto. Fiber boxes conforming to Specification PPP–B–636 may be banded with tape conforming to type IV of Specification PPP–T–97 and appendix thereto in lieu of steel straps. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.2 *Level B.* Shunts packaged as specified (see 6.2) shall be packed in domestic-type wirebound wood, wood-cleated fiberboard, wood-cleated plywood, nailed wood, fiber (class 1 or 2, as specified (see 6.2)), or wood-cleated paper-overlaid boxes conforming to Specifications PPP–B–585, PPP–B–591, PPP–B–601, PPP–B–621, PPP–B–636, and MIL–B–10377, respectively, at the option of the manufacturer. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.3 *Level C.* Shunts packaged as specified (see 6.2) shall be packed in containers of the type, size, and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Uniform Freight Classification Rules, or regulations of other carriers as applicable to the mode of transportation.

5.2.4 *General.* Insofar as possible and practical, exterior containers shall be uniform in shape and size, shall be of minimum cube and tare consistent with the protection required, and shall contain identical quantities of identical items.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and exterior shipping containers shall be marked in accordance with Standard MIL–STD–129 (see 6.2).

6. NOTES

6.1 Intended use. Shunts covered by this specification are intended for general use with panel-type, 50 mv instruments of approximately 2 percent accuracy.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Complete type designation (see 1.2.1 and 3.1).

(c) Whether additional identification marking on the shunt is required (see 3.9).

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(d) Laboratory at which product inspection is to be conducted (see 4.1.1).

(e) Levels of preservation and packaging, and applicable marking (see sec. 5).

(f) Whether shunt leads are required.

(g) Method of preservation and packaging of Specification MIL–P–116, if other than method IA (see 5.1.1).

(h) Class of fiber (see 5.2.1 and 5.2.2).

6.2.1 *Indirect shipments.* The packaging, packing, and marking specified in section 5 apply only to direct purchases by or direct shipments to the Government and are not intended to apply to contracts or orders between the manufacturer and prime contractor.

6.3 Application. When current of 100 amperes or greater is passing through a shunt, the major portion of the heat generated is dissipated by conduction through the shunt-terminal blocks into the connecting bus bar or cable. Therefore, it is necessary to insure that good contact is made between the shunt-terminal blocks and the conductor terminals, and that the conductors have adequate cross section and heat dissipation to keep the temperature of the shunt from exceeding 145°C. Good contact requires that the face and conductor-terminal tongue be flat. Sufficient pressure must be exerted so that the electrical resistance at the joint will be low to avoid the generation of excess heat, and so that good thermal conductivity will be obtained. In general, the contacting surface of the conductor terminals, as well as the contact faces and washers of the shunt, should be carefully cleaned before making the joint. Proper orientation of all lead terminals connected to the shunt is necessary to insure maximum accuracy of instrument readings (see 4.3.2).

6.3.1 *Mounting position.* For shunts of 10 ampere rating or greater, heat dissipation is an important consideration. For these shunts, the leaves should be vertical, with the longitudinal axis of the shunt in a horizontal position, in order to promote the free convectional flow of air. Where this mounting position is impractical and in installations where the shunt is in a confined location, forced air cooling should be provided.

6.3.2 *Shunt leads.* Instruments for use with external shunts are generally calibrated to include the resistance of the shunt leads normally supplied with the specific instrument. Therefore, when connecting these leads to the shunt, the length of the leads should not be altered in order to avoid introduction of error in measurement. If it is essential to increase the lead length, the wire size should be increased to a value which will provide the total loop resistance of the original value. Precautions should be taken, before attaching lead terminals to the shunts, to make sure that all contacting surfaces are clean and smooth. The lugs should be secured in a position which is parallel to current flow.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permision to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Signal Corps
Navy—Bureau of Ships
Air Force

Preparing activity:
Army—Signal Corps

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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